

Cracking Cocktail Parties: Neural Signatures of Auditory Processing



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Introduction

Cocktail Party Problem

- Everyday listening often requires focusing on one speaker while ignoring others.
- Temporal Response Functions (TRFs)** track how the brain processes speech, revealing both acoustic and higher-level linguistic processing.

What we know

- Our brain tracks attended speech, even when streams overlap.
- Spatial cues generally help source separation.

What we don't know

- How do different listening scenarios differ **functionally** in shaping brain responses?

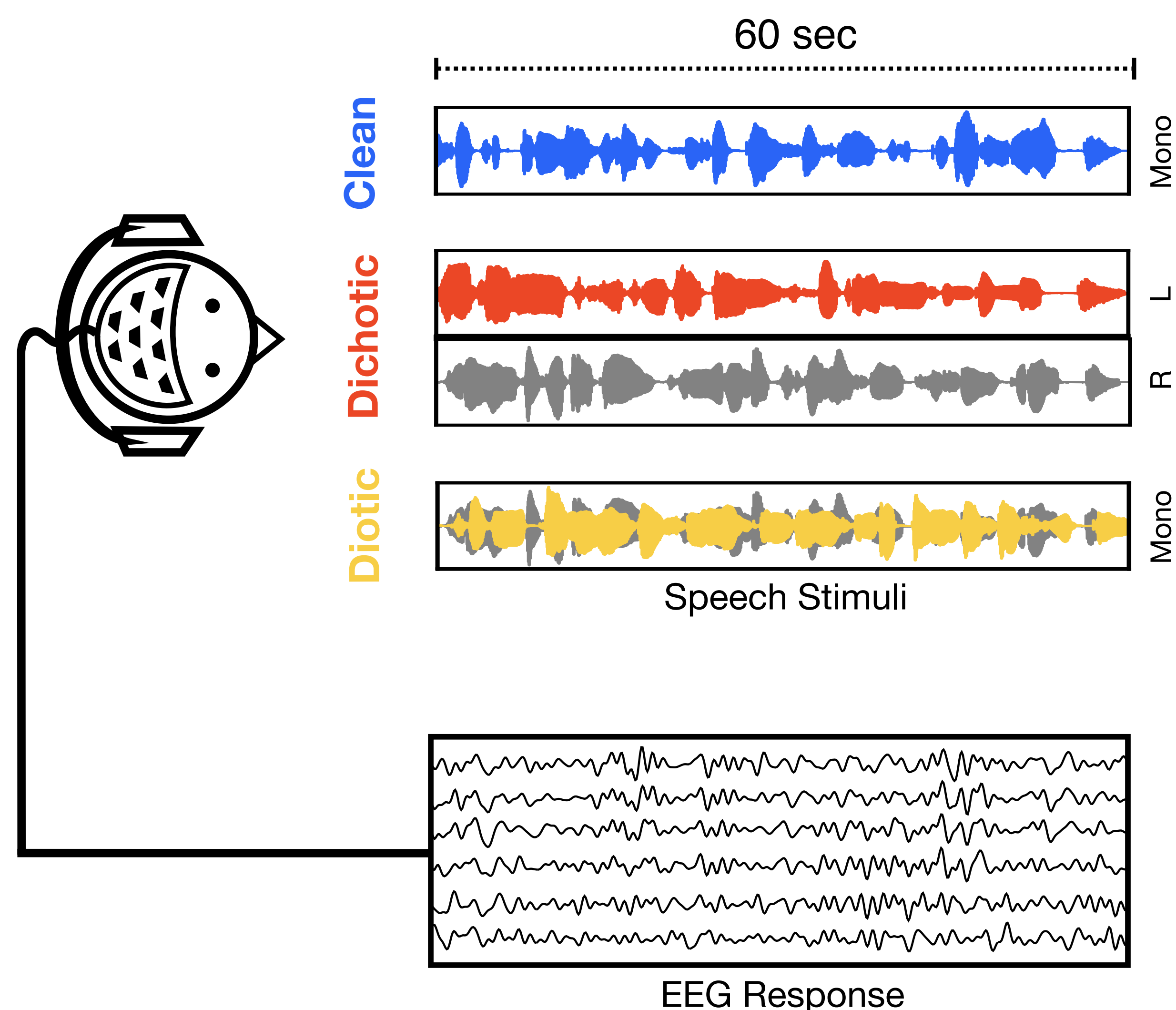
Our study

- Compared **TRFs** across **Clean**, **Dichotic**, and **Diotic** listening.
- Investigated how sensory and attentional demands alter neural signatures.

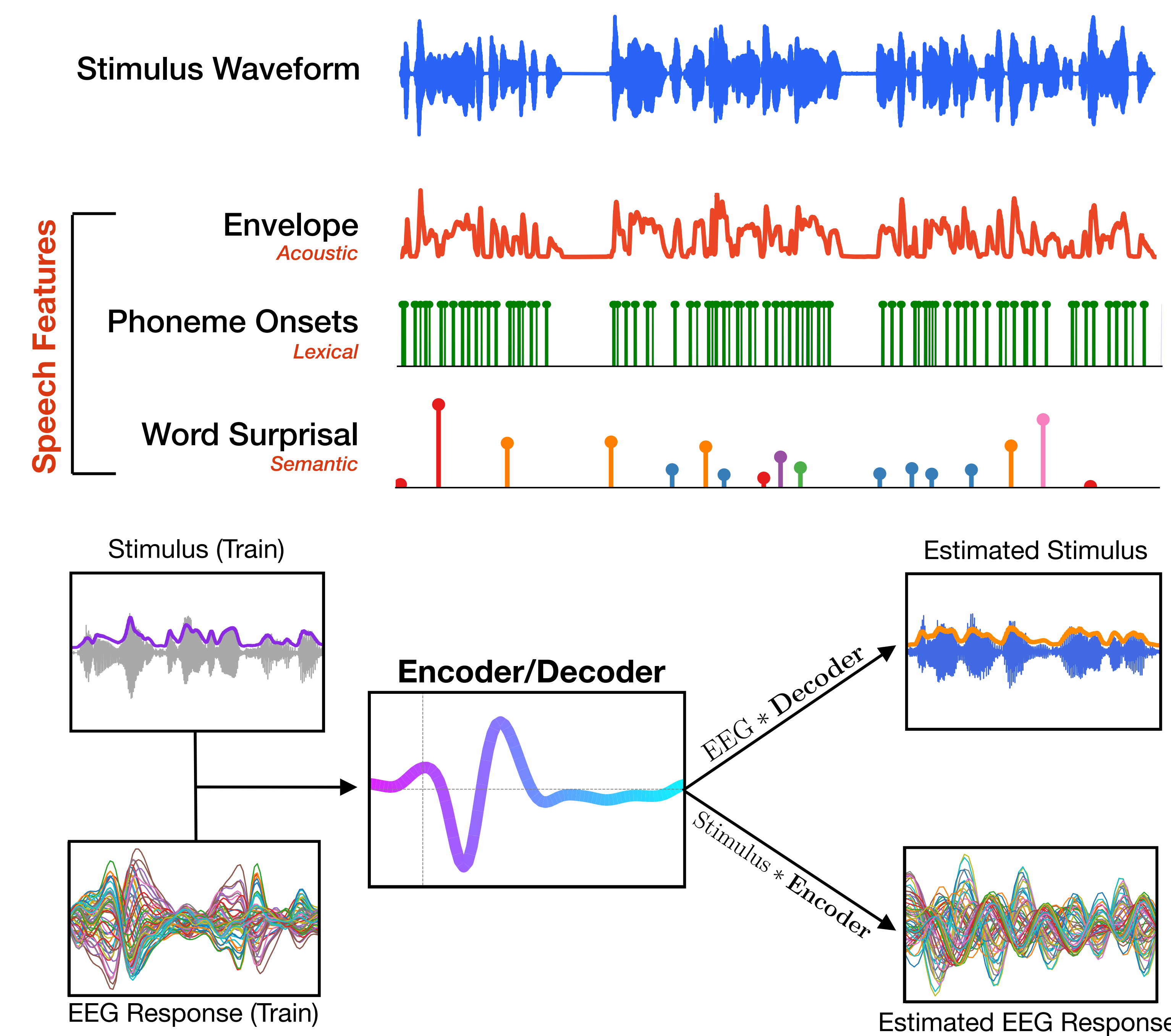
Hypotheses

Clean vs Diotic → Central separation demands modulate TRFs.

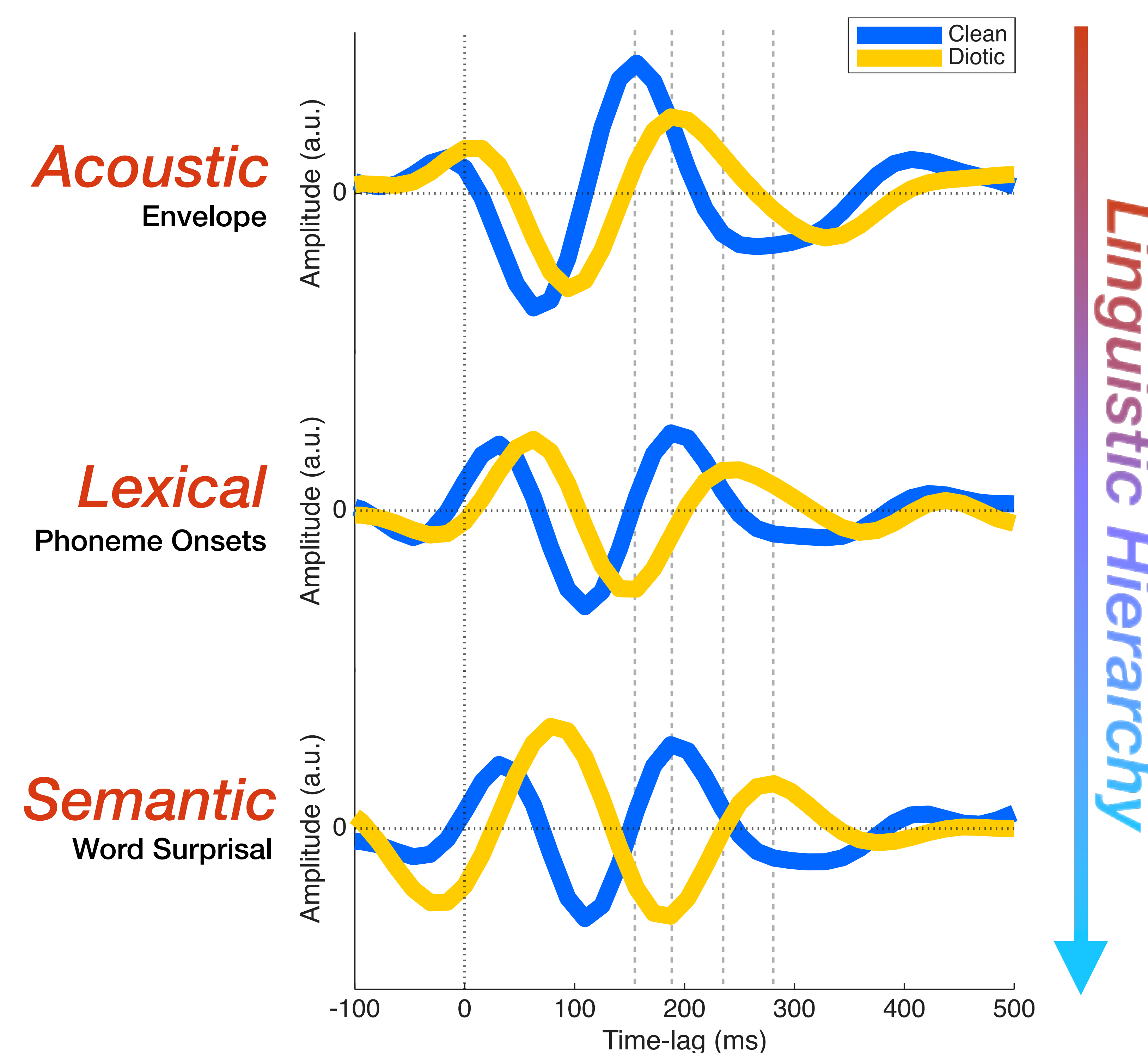
Dichotic vs Diotic → Spatial cues facilitate more efficient segregation.



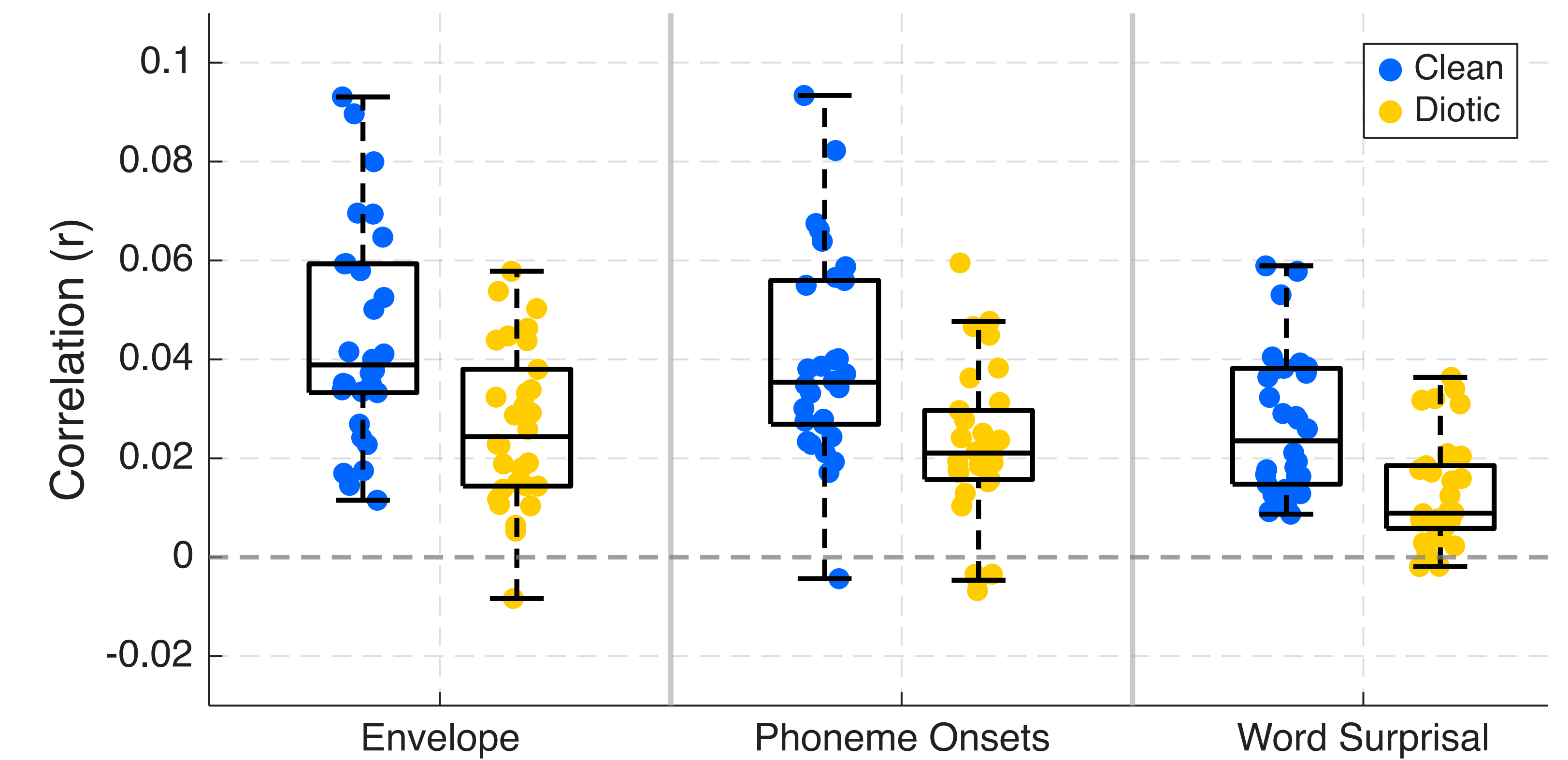
Methods



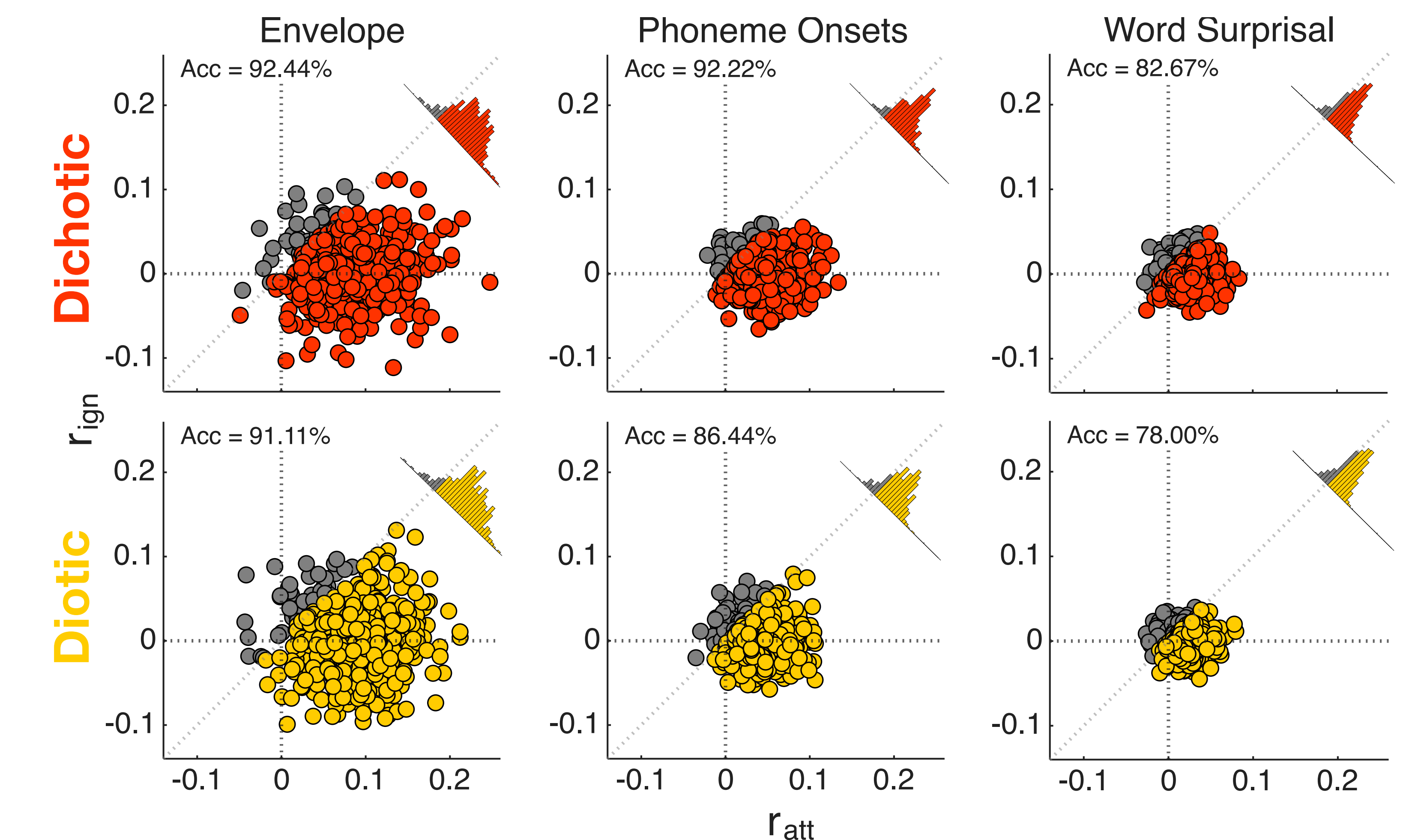
Results



Encoder Performance



Decoder Performance



Discussion

Clean vs Diotic

- TRF peaks (N1/P2):** Delayed in **Diotic** relative to **Clean**
→ Reflects processing load of central separation.
- Hierarchical latency gaps:** Stable in **Clean**, but expanded in **Diotic**
→ Central separation delays higher-level processing.

Dichotic vs Diotic

- Auditory attention decoding:** Better performance in **Dichotic**
→ Spatial cues enable faster, clearer target representations.

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (Ministry of Science and ICT) (RS-2024-00461617)